INSPIRE FORENSICS

(Recognised by the Indian Society of Forensic Science)

CUET-UG 2026

SAMPLE PAPER

A National Level Socio-Forensic Voluntary Student-Led Educational Firm

PHYSICS (322)

1) Out of gravitational, electromagnetic, Vander Waals, electrostatic and nuclear forces; which two are able to provide an attractive force between two neutrons

a. Electrostatic and gravitational

b. Electrostatic and nuclear

c. Gravitational and nuclear

d. Some other forces like Vander Waals

2) The dielectric constant of pure water is 81. Its permittivity will be:

a. 7.12 * 10 ^ - 10 MKS units

b. 8.86 * 10 ^ - 12 MKS units

c. 1.02 * 10 ^ 13 MKS units

d. Cannot be calculated

3) Three charges 4q, Q, and q are in a straight line in the position of 0, l/2, and l respectively. The resultant force on q will be zero if Q equal to:

a. -q

b. -2q

c. -q/2

d. 4q

4) In optical fibre, the light travel along the fibre by undergoing repeated total internal reflections. For this purpose, we ensure that the refractive index of the material:

a. is low

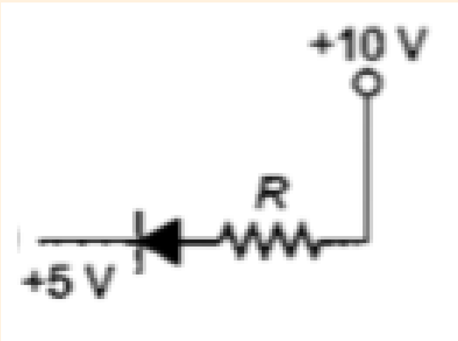
b. decreases from centre towards surface

c. increases from centre towards surface

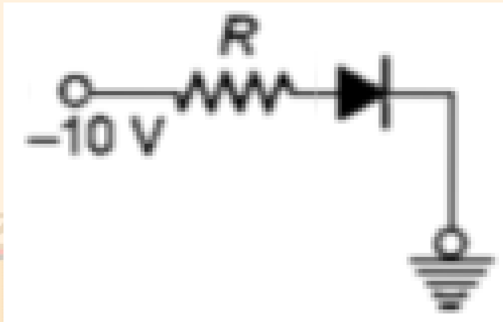
d. is high

5) In the following figure, the diodes which are forward biased, are

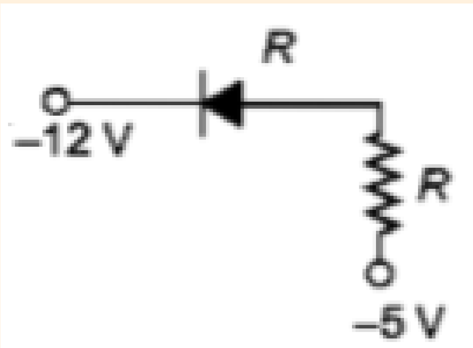
(I)



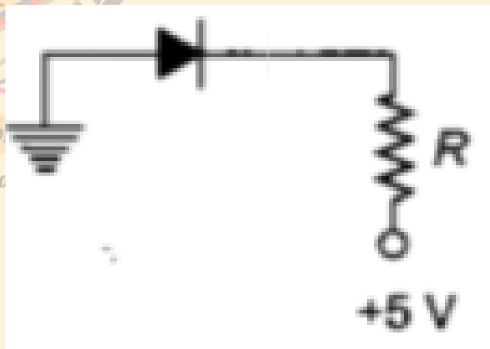
(II)



(III)



(IV)



Choose the correct option from the given ones:

a. III and IV only

b. I and III only

c. II only

d. II and IV only

6) In forward biasing of the p-n junction

a. the positive terminal of the battery is connected to n-side and the depletion region becomes thin

b. the positive terminal of the battery is connected to n-side and the depletion region becomes thick

c. the positive terminal of the battery is connected to p-side and the depletion region becomes thin

d. the positive terminal of the battery is connected to p-side and the depletion region becomes thick

7) When a p-n junction diode is reversed bias, the width of the depletion region,

a. Increases

b. Decreases

c. Remains unaltered

d. Becomes zero

8) What is the respective number of a and B-particles emitted in the following radioactive decay?

$^{200}X_{90}$ **-->** $^{168}Y_{80}$

a. 6 and 8

b. 6 and 6

c. 8 and 8

d. 8 and 6

9) A free neutron decays into a proton, an electron and:

a. A beta particle

b. An alpha particle

c. An antineutrino

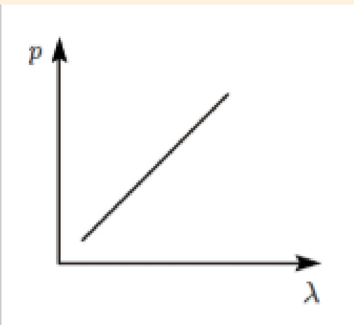
d. A neutrino

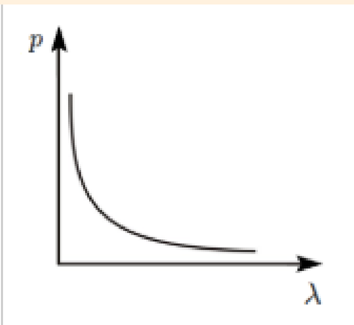
10) The activity of a radioactive sample is measured as 9750 counts/min at $t = 0$ and as 975 counts/min at $t = 5$ min. The decay constant is approximately:

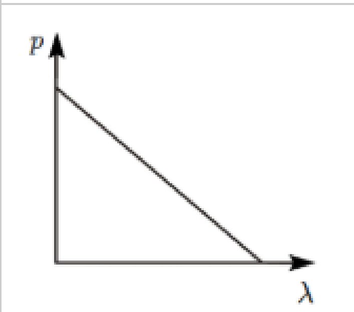
a. 0.922/min
b. 0.691/min
c. 0.461/min
d. 0.230/min

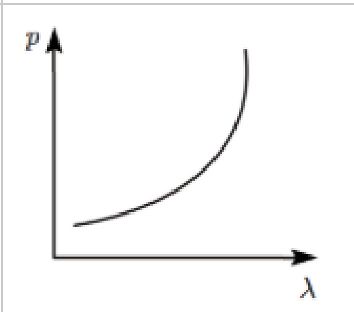
11) Consider the spectral line resulting from transition $n = 2$ to $n = 1$ in the atoms and ions given below. The shortest wavelength is given by:
a. hydrogen atom
b. deuterium
c. singly ionised helium
d. doubly ionised lithium

12) Which of the following figures represents the variation of the particle momentum and the associated de-Broglie wavelength?

a. 

b. 

c. 

d. 

13) A 200W sodium street lamp emits yellow light of wavelength 0.6 μm. Assuming it to be 25% efficient in converting electrical energy to light, the number of photons of yellow light it emits per second is

a. 1.5 * 10 ^ 20

b. 6 * 10 ^18

c. 62 * 10 ^ 20

d. 3 * 10 ^ 19

14) The rest mass of the photon is

a. 0

b. ∞

c. Between 0 and ∞

d. Equal to that of an electron

15) In a double-slit experiment, the two slits are 1 mm apart and the screen is placed 1 m away. A monochromatic light of wavelength 500 nm is used. What will be the width of each slit for obtaining ten maxima of double-slit within the central maxima of a single-slit pattern?

a. 0.2 mm

b. 0.1 mm

c. 0.5 mm

d. 0.02 mm

16) Two waves of intensity ratio 9:1 interfere to produce fringes in a young's double-slit experiment, the ratio of intensity at maxima to the intensity at minima is

a. 4:1

b. 9:1

c. 81:1

d. 9:4

17) The ratio of the amplitude of the magnetic field to the amplitude of electric field for an electromagnetic wave propagating in vacuum is equal to
a. the speed of light in vacuum
b. reciprocal of speed of light in vacuum
c. the ratio of magnetic permeability to the electric susceptibility of vacuum
d. unity

18) The decreasing order of the wavelength of infrared, microwave, ultraviolet and gamma rays is
a. gamma rays, ultraviolet, infrared, microwaves
b. microwaves, gamma rays, infrared, ultraviolet
c. infrared, microwave, ultraviolet, gamma rays
d. microwave, infrared, ultraviolet, gamma rays

19) Which of the following statements is false for the properties of electromagnetic waves?
a. Both electric and magnetic field vectors attain the maxima and minima at the same place and the same time
b. The energy in an electromagnetic wave is divided equally between electric and magnetic vectors
c. Both electric and magnetic field vectors are parallel to each other and perpendicular to the direction of propagation of the wave
d. These waves do not require any material medium for propagation

20) When 100 volt d.c. is applied across a solenoid, a current of 1.0 A flows in it. When 100 volt a.c. is applied across the same coil, the current drops to 0.5 A. If the frequency of a.c. source is 50 Hz the impedance and inductance of the solenoid is
a. 200 Ω and 0.55 H
b. 100 Ω and 0.86 H
c. 200 Ω and 1.0 H
d. 100 Ω and 0.93 H

21) A direct current of 2 A and an alternating current having peak value 2 A flow through two identical resistances at the same time. The ratio of heat produced in the two resistance will be:

a. 1:1

b. 1:2

c. 2:1

d. 4:1

22) The maxwell's equation:

$$\oint \overrightarrow{B}.\overrightarrow{dl} = \mu_0\left(i + \varepsilon_0.\frac{d\phi_E}{dt}\right)$$

is a statement of:

a. Faraday's law of induction

b. Modified Ampere's law

c. Gauss's law of electricity

d. Gauss's law of magnetism

23) Which of the following is not based on the application of eddy currents?

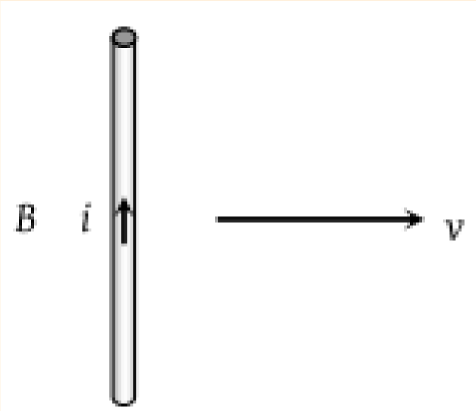
a. Speedometer

b. Dead-beat galvanometer

c. Energy meter

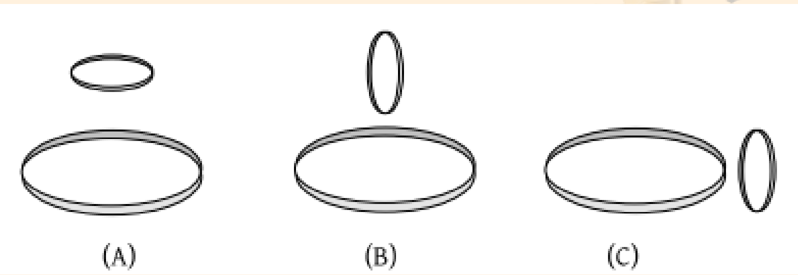
d. Tangent galvanometer

24) A conducting wire is moving towards right in a magnetic field B. The direction of induced current in the wire is shown in the figure. The direction of magnetic field will be:



a. In the plane of paper pointing towards right
b. In the plane of paper pointing towards left
c. Perpendicular to the plane of paper and down wards
d. Perpendicular to the plane of paper and upwards

25) Two circular coils can be arranged in any of the three situations shown in the figure. Their mutual inductance will be



a. Maximum in situation (A)
b. Maximum in situation (B)
c. Maximum in situation (C)
d. The same in all situations

26) In an LR-circuit, the time constant is that time in which current grows from zero to the value (where Io is the steady-state current)

a. 0.63 Io

b. 0.50 Io

c. 0.37 Io

d. Io

27) Two small bar magnets are placed in the air at a distance r apart. The magnetic force between them is proportional to:

a. r^2

b. r^{-2}

c. r^{-3}

d. r^{-4}

28) The magnetic susceptibility of a substance is -0.0002 at room temperature. If this substance is kept in a uniform external magnetic field B, the net field inside the substance becomes B', then:

a. B'>B

b. B'<B

c. B' = B

d. B' = 0

29) For the magnetic field to be maximum due to a small element of current-carrying conductor at a point, the angle between the element and the line joining the element to the given point must be:

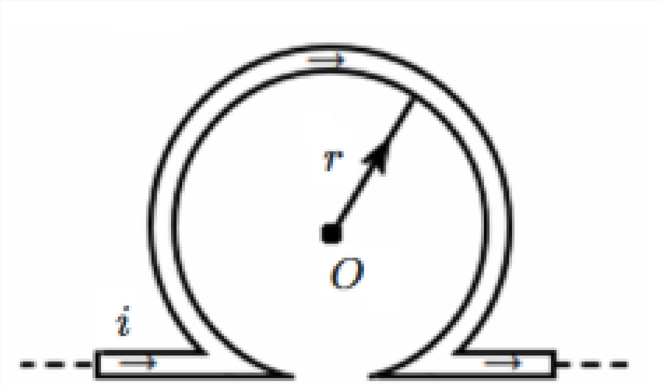
a. 0°

b. 90°

c. 180°

d. 45°

30) An infinitely long straight conductor is bent into the shape as shown in the figure. It carries a current of i amperes and the radius of the circular loop is r metres. What will be the magnetic induction at its



a. μo 2i (π + 1) / 4π r

b. μo 2i (π - 1) / 4π r

c. zero

d. Infinite

31) A uniform electric field and a uniform magnetic field are acting in the same direction in a certain region. If a proton is projected in this region such that its velocity is pointed along the direction of fields, then the proton:

a. will turn towards the left of the direction of motion

b. will turn towards the right of the direction of motion

c. will speed up

d. will slow down

32) Only the current inside the Amperian loop contributes in:

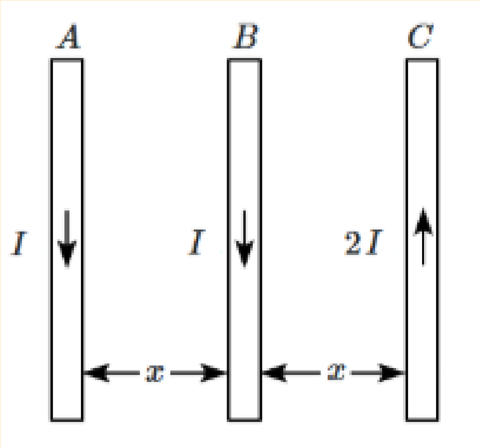
a. finding magnetic field at any point on the Ampere's loop

b. line integral of magnetic field

c. in both of the above

d. in neither of them

33) A, B, and C are parallel conductors of equal length carrying currents I, I, and 2I respectively. The distance between A and B is x. The distance between B and C is also x. F_1 is the force exerted by B on A and F_2 is the force exerted by C on A choose the correct answer:



a. $F_1 = 2F_2$
b. $F_2 = 2F_1$
c. $F_1 = F_2$
d. $F_1 = -F_2$

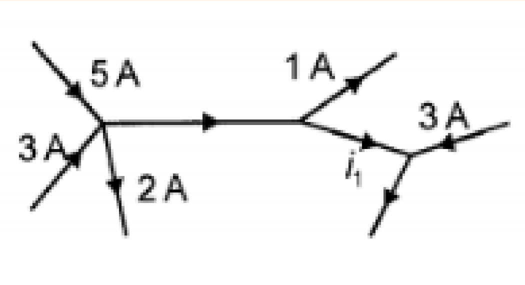
34) The magnetic moment of a current (i) carrying circular coil of radius (r) and number of turns (n) varies as:
a. $1/r^2$
b. $1/r$
c. r
d. r^2

35) The resistivity of a wire:
a. Increases with the length of the wire
b. Decreases with the area of cross-section
c. Decreases with the length and increases with the cross-section of the wire
d. None of the above statement is correct

36) The positive temperature coefficient of resistance is for:
a. Carbon
b. Germanium
c. Copper
d. An electrolyte

37) For driving a current of 2 A for 6 minutes in a circuit, 1000 J of work is to be done. The e.m.f. of the source in the circuit is
a. 1.38 V
b. 1.68 V
c. 2.04 V
d. 3.10 V

38) The value of electric current i, in the network below is



a. 5A
b. 9A
c. 8A
d. 7A

39) A battery has emf 4 V and internal resistance r. When this battery is connected to an external resistance of 2 ohm, a current of 1 ampere flows in the circuit. How much current will flow if the terminals of the battery are connected directly?
a. 1A
b. 2A
c. 4A
d. Infinite

40) Four identical cells each having an electromotive force (e.m.f.) of 12V, are connected in parallel. The resultant electromotive force (e.m.f.) of the combination is:

a. 48 V

b. 12 V

c. 4 V

d. 3 V

41) A voltmeter has resistance of 2000 ohms and it can measure upto 2V. If we want to increase its range to 10 V, then the required resistance in series will be

a. 2000 Ω

b. 4000 Ω

c.6000 Ω

d.8000 Ω

42) A ray of light traveling in a transparent medium of refractive index μ falls, on a surface separating the medium from the air at an angle of incidence of 45°. For which of the following values of μ the ray can undergo total internal reflection?

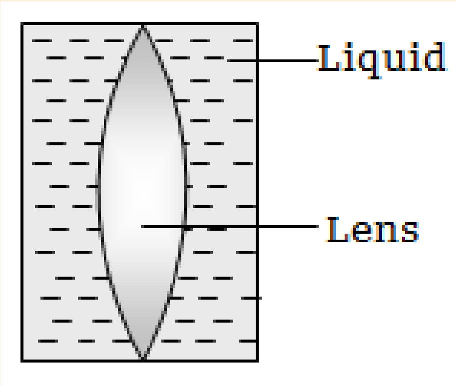
a. μ = 1.33

b. μ = 1.4

c. μ = 1.5

d. μ = 1.25

43) Shown in the figure here is a convergent lens placed inside a cell filled with a liquid. The lens has focal length + 20 cm when in air and its material has refractive index 1.50. If the liquid has refractive index 1.60, the focal length of the system is



a. + 80 cm
b. -80 cm
c. -24 cm
d. -160 cm

44) Two similar thin equi-convex lenses, of focal length f each, are kept coaxially in contact with each other such that the focal length of the combination is F1. When the space between the two lenses is filled with glycerin which has the same refractive index as that of glass (mu = 1.5) then the equivalent focal length is F2. The ratio F_1: F_2 will be:
a. 3:4
b. 2:1
c. 1:2
d. 2:3

45) An object is placed at a distance of 40 cm from a concave mirror of a focal length of 15 cm. If the object is displaced through a distance of 20 cm towards the mirror, the displacement of the image will be:
a. 30 cm away from the mirror.
b. 36 cm away from the mirror.
c. 30 cm towards the mirror.
d. 36 cm towards the mirror.

46) The electric potential V at any point O (x, y, z all in metres) in space is given by $V = 4x^2$ volt. The electric field at the point (1m, 0, 2m) in volt/metre is-

a. 8 along negative x-axis

b. 8 along positive x-axis

c. 16 along negative x-axis

d. 16 along positive z-axis

47) An uncharged aluminium block has a cavity within it. The block is placed in a region permeated by a uniform electric field which is directed upwards. Which of the following is a correct statement describing conditions in the interior of the block's cavity?

a. The electric field in the cavity is directed upwards

b. The electric field in the cavity is directed downwards

c. There is no electric field in the cavity

d. The electric field in the cavity is of varying magnitude and is zero at the exact center.

48) A parallel plate condenser is immersed in an oil of dielectric constant 2. The field between the plates

a. Increased proportional to 2

b. Decreased proportional to 1/2

c. Increased proportional to √2

d. Decreased proportional to 1/√2

49) Two spheres of radius a and b respectively are charged and joined by a wire. The ratio of the electric field at the surface of the spheres is

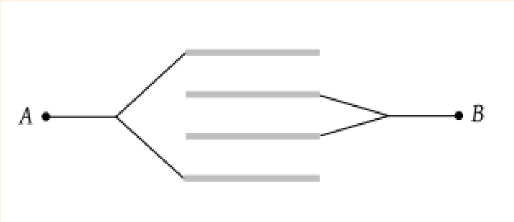
a. a/b

b. b/a

c. a^2/b^2

d. b^2/a^2

50) Four plates of equal area A are separated by equal distances d and are arranged as shown in the figure. The equivalent capacity is



a. 2εoA/d
b. 3εoA/d
c. 4εoA /d
d. εoA/d

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Answer Key

Question number	Answer Key		Question number	Answer Key
1	C		19	C
2	A		20	A
3	A		21	C
4	B		22	B
5	C		23	D
6	C		24	D
7	A		25	A
8	D		26	A
9	C		27	D
10	C		28	B
11	D		29	B
12	D		30	C
13	A		31	C
14	A		32	B
15	A		33	D
16	A		34	D
17	B		35	D
18	D		36	C

Answer Key

Question number	Answer Key
37	A
38	D
39	B
40	B
41	D
42	C
43	D
44	C
45	B
46	A
47	C
48	B
49	B
50	A